LAB 6 – CODE COVERAGE ANALYSIS AND REPORT

Experiment: Code Coverage Analysis of Bus Transit Fare Application
Application: bus_transit_fare.py
Test File: CodeCoverageTest_PES2UG24CS570_application.py

1. Objective

To analyse the code coverage of the Bus Transit Fare application using Python's Coverage.py tool and pytest. Additional test cases were developed to ensure that all executable statements in the application were covered during testing.

2. Introduction

Code coverage measures the extent to which the source code is executed when the test cases are run. It helps identify statements that are not executed during testing.

For this experiment, statement coverage was measured using the Python Coverage.py library. The aim was to achieve at least 90% code coverage.

3. Test Case Development

Additional test cases were written to test different functionalities and conditions of the Bus Transit Fare application.

The test cases covered:

- Invalid manifest input and invalid booking fee.
- Different route types, including express, shuttle and local.
- Different passenger categories, including regular passengers, students and senior citizens.
- Priority boarding and booking fee calculations.
- Inactive routes and invalid commuter zones.
- Blacklisted commuters.
- Rejected rides and audit conditions.
- Zero-ridership conditions.
- Smartcard turnstile inspection, including operational and jam conditions.

A total of 9 test cases were executed successfully.

4. Execution

The following commands were used to execute the test cases and generate the code coverage report:

python -m coverage erase

python -m coverage run --include="bus_transit_fare.py" -m pytest CodeCoverageTest_PES2UG24CS570_application.py

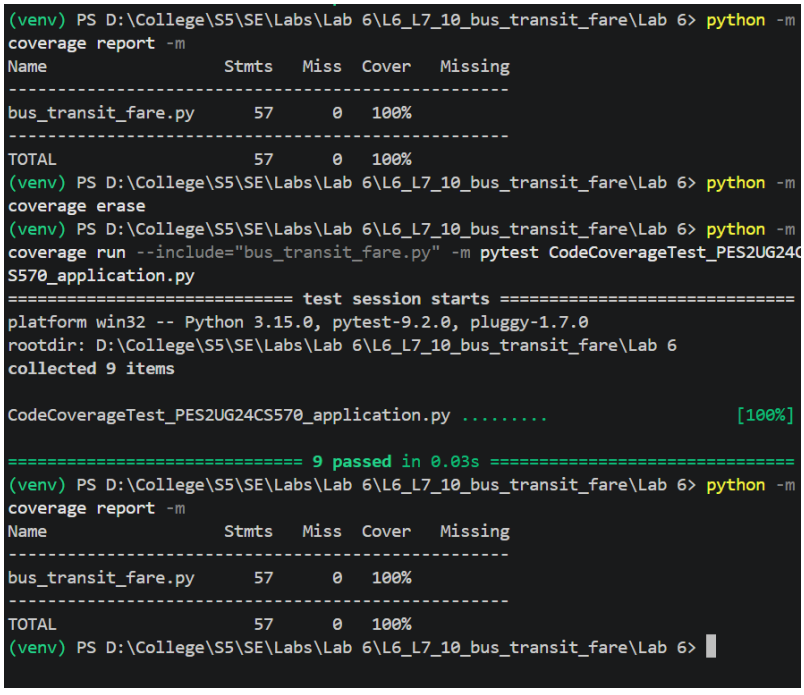
python -m coverage report -m

5. Results and Analysis

The test execution completed successfully with all 9 test cases passing.

Metric	Result
Total statements	57
Missing statements	0
Code coverage	100%

The coverage report showed that all 57 executable statements in bus_transit_fare.py were executed by the test cases. No statements remained uncovered.



6. Conclusion

Code coverage analysis was successfully performed on the Bus Transit Fare application using pytest and Coverage.py. The additional test cases achieved 100% code coverage, exceeding the required minimum of 90%. This indicates that all executable statements in the application were exercised during testing.